

Engineering Report: Solutions to a cleaner planet

Optimising solar panels in the MENA region for improved efficiency

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Executive Summary

The 7th Sustainable Development Goal (SDG) aims to ensure clean, stable, and economical energy access for all by 2030 [1].

This report explores an innovative solution that can directly lead to meaningful progress towards this goal by increasing the efficiency of solar panels through the installation of independent self-cleaning systems powered by Savonius wind turbines and thermoelectric Peltier modules, along with specialised sensors and artificial intelligence-based systems.

The Engineering Solution

According to Adekanbi et al. (2023), soiling from dust and dirt can significantly reduce the efficiency of solar panels in arid regions such as the Middle East and North Africa, with estimated efficiency drops of up to 40% after 2 months of soiling. [2]. To address this, an automated solar-panel cleaning system that eliminates dependence on external water resources is proposed. This system operates by combining Savonius wind turbines with thermoelectric atmospheric water generators (TEAWG). The Savonius turbine transforms wind energy into rotational mechanical energy, which is then converted to electrical energy via a generator. This electrical energy powers the thermoelectric Peltier modules. These modules cool a condenser surface below the dew-point temperature of the ambient air, causing water vapour to condense into liquid water. Next, the liquid water is collected into a storage tank where it can be stored until future use. When cleaning is required, a pump or gravity-based system can force the water to flow over the solar panel surfaces to remove accumulated dust and dirt. Additionally, IoT sensors, such as pyranometers and

photovoltaic (PV) panel performance sensors, can be used to monitor solar irradiance and panel output, allowing artificial intelligence (AI)-based control systems to estimate soiling and automatically activate cleaning when required. A comparable self-cleaning approach without external water input demonstrated a 28% increase in energy output in published testing by Omar et al. [3].

A crucial component of this system is the Savonius wind turbine. This type of turbine was chosen because many locations in the MENA region experience relatively low wind speeds, which may limit the performance of conventional wind turbines. Instead, Savonius turbines are not only simple in design and require low manufacturing costs, but they have strong self-starting capability, making them well suited for MENA regions where wind speeds are low [4]. Additionally, this type of turbine is commonly found in a vertical-axis setup and can also be referred to as a VAWT (vertical-axis wind turbine). A benefit of such a setup is that they can begin turning regardless of wind direction.

The figure below illustrates a brief concept of the 5 key steps involved in the proposed system:



Figure 1: The steps proposed for self-cleaning solar panels

Analysing Feasibility and Limitations

In order to simplify the analysis of real-world feasibility of this solution, the following educated assumptions will be made by considering the functionality of the system during summer periods across the United Arab Emirates only:

- 1) In coastal areas of the United Arab Emirates (UAE), average summer humidity ranges between 60 to 90 percent. We will therefore assume an average humidity value to be in the midpoint of these figures; 75%.
- 2) Average wind speeds during the summer in the UAE range from between 10 km/h to 16 km/h. Taking the average of these values, we will assume average wind speed to be 13 km/h, equivalent to roughly 3.6 m/s.
- 3) The most common thermoelectric Peltier modules will be used, which are the TEC112706 modules.
- 4) Summer temperatures in the UAE often vary between 35 and 45°C. For this analysis, we will assume the average temperature to be 38°C.

The above data can be summarised to:

Relative humidity (%)	Average wind speed (<i>m/s</i>)	Average temperature (°C)
75	3.6	38

A published study investigating the relationship between humidity, ambient temperature and the condensation rate produced by the Peltier modules [5] will be used with the above assumptions. Using the published data, we can conclude the following:

When an average humidity level of 60% was observed, a graph of the condensation rate in mL/h against the temperature in degrees Celsius showed an approximately linear relationship between the two variables. By mathematically analysing this graph, we can estimate that a change 1°C increases the condensation rate by about 1.3 to 1.5 mL/h. Therefore, at our assumed average temperature of 38°C, we can predict the condensation rate to be about 26 mL/h. However, this result still assumes an average humidity value of 60%, which is far from our average assumed humidity of 75%. Since no published condensation-rate data were available at 75% relative humidity, a simplifying assumption is made that the relationship between the condensation rate and relative humidity is approximately proportional, since higher humidity directly increases the mass of water vapour available for condensation. Applying this proportionality to the known value of 26 mL/h at 60% humidity gives an estimated condensation rate at our assumed relative

humidity of 75% of: $26 \times \left(\frac{75}{60}\right) = 32.5$ mL/h. For this analysis, a conservative estimate of 32 mL/h is used. It is acknowledged that this proportionality is a simplification, since condensation behaviour may become non-linear at higher humidity levels closer to saturation. However, in the absence of published data at 75% humidity, this provides a reasonable estimate for this feasibility analysis.

Next, we can find the dew point of water in conditions with a temperature of 38°C and a relative humidity of 75%. Using the Magnus-Tetens formula, and assuming the accepted constants of a and b to be 17.27 and 237.7°C respectively, we apply the equation to get a dew point of roughly 33°C. This shows that in order to condense surrounding water vapour at our conditions, we need to cool air to a temperature below 33°C. However, a large temperature delta between the dew point of 33°C and the temperature of the cool surface on the Peltier unit will help force condensation due to a significant thermal gradient. As a result, this may require us to maintain a stable temperature on the cold side of the Peltier module of 20 to 25°C.

The magnitude of power generated by the Savonius wind turbine can be approximated by the formula:

$\frac{1}{2}\rho Av^3$, where v is 3.6 meters per second and A is the swept area of each turbine, which can be calculated by the product of the turbine's rotor height, H , and diameter, D ($A = H \times D$).

However, the turbine cannot be 100% efficient, so by adding a factor of power coefficient to our value of about 15%, which is a value typically associated with Savonius turbines, we get a new expression for expected power to be $\frac{3}{40}\rho Av^3$. This result highlights an important point and a significant limitation of the system: Savonius wind turbines feature very low power coefficients, making them highly inefficient when compared to other types of wind turbines.

Regardless, if we assume this solution were to be implemented on a large scale in solar farms where the farm would feature an array of 100 solar panels, we can consider several of these turbines will be used to generate power to the Peltier modules, instead of only a few extremely large turbines. Therefore, we can find the required swept area of each turbine using the data we have calculated, in the following way:

- 1) Assuming each of the 100 panels will use 0.5 liters of water to effectively clean their surfaces, we will require a total of $100 \times 0.5 = 50$ liters of water per cleaning cycle.

- 2) Assuming that solar panels in dusty environments require cleaning every 2 to 4 weeks, we have at least 2 weeks (14 days) to collect the 50 liters of water. This means that we require condensation and storage of water at a rate of $\frac{50}{14 \times 24} = 0.1488 \text{ L/h}$ (equivalent to 148.8 mL/h).
- 3) The Peltier modules used (TEC1-12706) are rated at a nominal voltage of 12V and a current of 3 to 6 amperes, although operating current is likely to be in the range of 4 to 5 amperes. However, for this analysis, we will assume maximum current, and therefore maximum power drawn, and can assume they will require a conservative estimate of $12 \times 6 = 72 \text{ W}$ of power in order to achieve our condensation rate. It may be worth noting that while real-world testing may suggest that this power estimate of the Peltier modules is significantly higher than what would be realistically required, it could account for the additional appliances such as cooling fans required to cool down and dissipate heat produced on the hot end of the Peltier modules.
- 4) To have 148.8 mL/h produced, we require a power value of $72 \times \frac{148.8}{32} = 335 \text{ W}$ generated by the turbines.
- 5) In practice, no system is 100% efficient and therefore energy losses will occur in the system due to inefficiencies, wiring losses, power electronics, and cooling equipment. Therefore, an overall system efficiency of 80% is assumed in this step. To provide the required 335 W to the Peltier modules, the turbines must therefore generate $\frac{335}{0.8} = 419 \text{ W}$. By equating this value of 419 W to our expected power expression of $\frac{3}{40} \rho A v^3$, where ρ can be assumed to be 1.15 kg/m^3 in the specific conditions, we reach a final value of the aggregated swept area of all turbines used to be approximately 104 m^2 .

Conclusion

In order to turn this concept into a feasible solution that can be implemented at scale in solar farms with 100 panels, we will require a total swept area across all turbines present of 104 m^2 , meaning the swept area of each turbine multiplied by the number of turbines must equal 104 m^2 . This figure does appear to be feasible, but it is important to note the large number of assumptions we have made to reach this value, including the unrealistic assumption that all power generated by the wind turbines will be used to power the Peltier modules. One last consideration that remains largely estimated is the amount of power that will be drawn by the Peltier modules. At already high ambient temperatures of between 35 to 45°C , the hot end of the module will likely have to stay at around at least 55°C .

Therefore, to dissipate this high heat, cooling fans will have to be used, which could increase the overall power drawn by the modules and therefore affect all subsequent calculations we have made after that point. For example, if we find that the power drawn by the Peltier units increases by 20% and now requires 86.4 W of power drawn, instead of the previous 72 W, we will have a new required aggregated swept area of approximately 125 m². However, until more detailed real-world testing is carried out, this solution appears to remain feasible at a large scale. Lastly, the mentioned dimensions of the wind turbines can be easily adjustable towards designer's specification if deemed impractical.

Appendix A: Parts List for the Automated System

The following table lists the key required components that make up the proposed concept.

Component	Purpose
Savonius wind turbine	Generate electricity
Wind turbine generator	Convert the rotational movement to electrical energy
TEC1-12706 Peltier modules	Condense the atmospheric moisture to form water droplets
Cooling fans	Dissipate heat from the hot ends of the Peltier modules
Storage tank	Store the liquid water droplets
Water pump	Pump the collected water through the pipes or hoses
Nozzles	Spray the water over the solar panel surfaces
Hoses	Allow the transportation of the water between 2 points
Pyranometer	Detect a buildup of grime of the surface
Temperature/humidity sensors	Monitor the external climate conditions
Microcontroller	Control the automated system
Battery storage system	Store the generated electrical energy

Appendix B: Savonius wind turbine design

The Savonius turbine was modelled using Fusion 360 to verify structural feasibility and assess the footprint requirements for the proposed system. The design focuses on a vertical-axis orientation to ensure consistent operation regardless of wind direction.

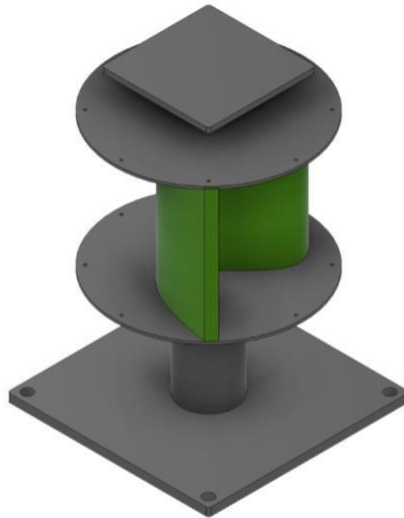


Figure 2: An isometric view of the modelled turbine

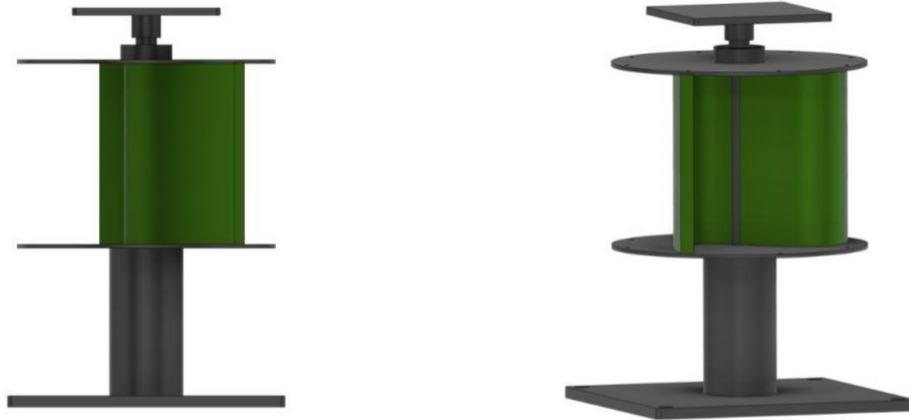


Figure 3 and 4: A side view of the modelled turbine

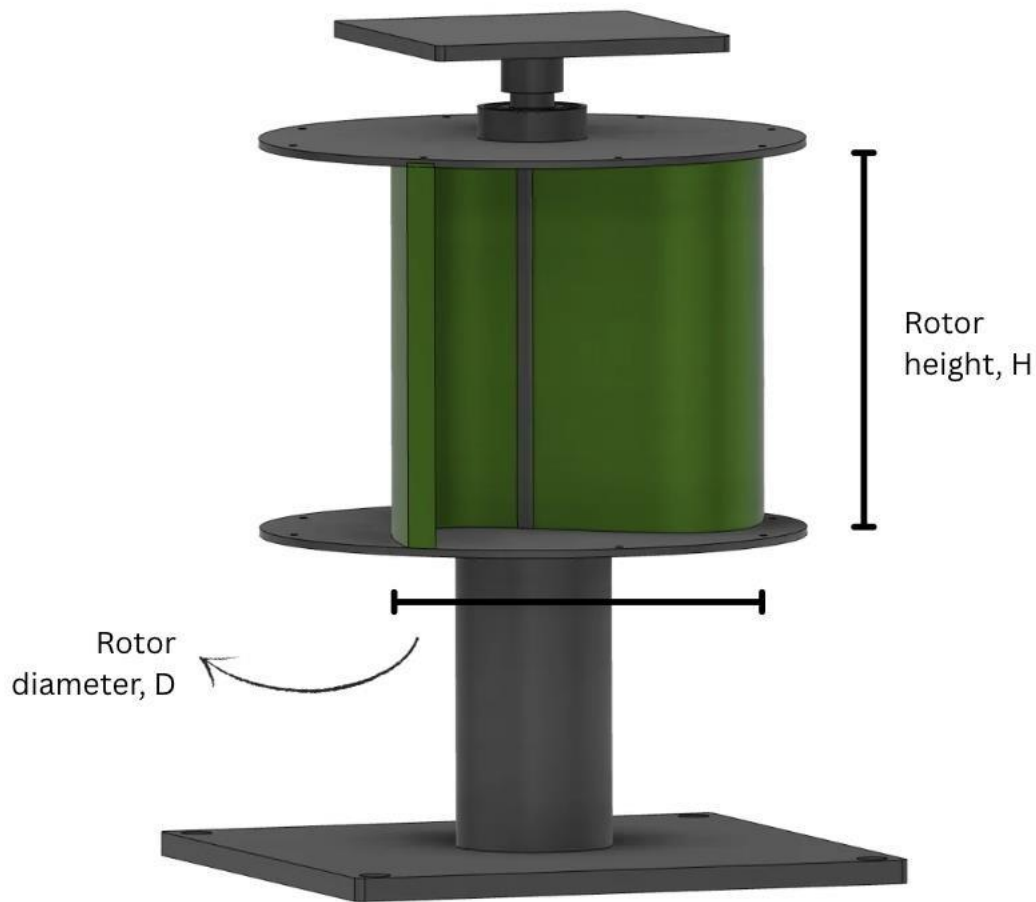


Figure 5:

A side view of the turbine, showing the marked dimensions of the rotor height and diameter. By multiplying the values of D and H , we can calculate the swept area of each turbine. There are infinitely many different combinations of rotor height, H , rotor diameter, D , and the number of turbines, n , we can use to reach the calculated total swept area required of 104 m^2 . For example, we could have a single Savonius turbine with a diameter of 1 meter and a height of 104 meters to theoretically generate the power to run the system.

However, this would certainly be unrealistic and impractical. Therefore, a smarter approach could be, for example, using 8 smaller Savonius turbines, each with a rotor diameter of 2 meters and a rotor height of approximately 6.5 meters.

References

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